Smart Car With Obstacle Detection

The Smart Car Which Can Find It's Own Way

Presented By

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Introduction

- Automated driving and obstacle detection are key advancements in vehicle safety.
- Our project aims to develop a robot that detects obstacles and adjusts its path using Arduino and sensors.
- The goal is to simulate a simplified version of real-world autonomous driving systems.

Objectives

- ✓ To simulate basic autonomous driving features such as obstacle avoidance and path adjustment.
- ✓ Detect obstacles accurately within a defined range.
- ✓ Automatically adjust the car's path to avoid collisions.

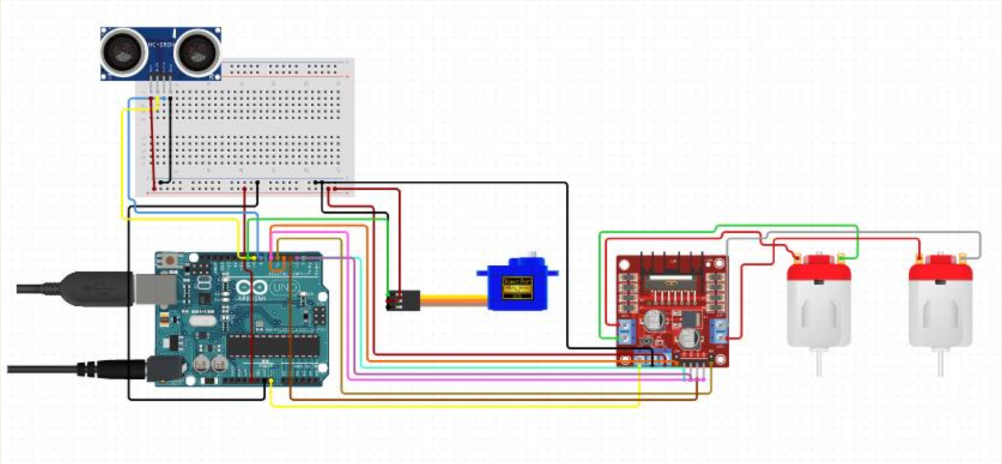
Relevance and scope

- Obstacle detection is crucial for self-driving cars, robots, and automated systems.
- Provides a framework for building more advanced autonomous systems.
- Smart Transport Applications: Highlights the potential of automation in enhancing road safety and traffic management.

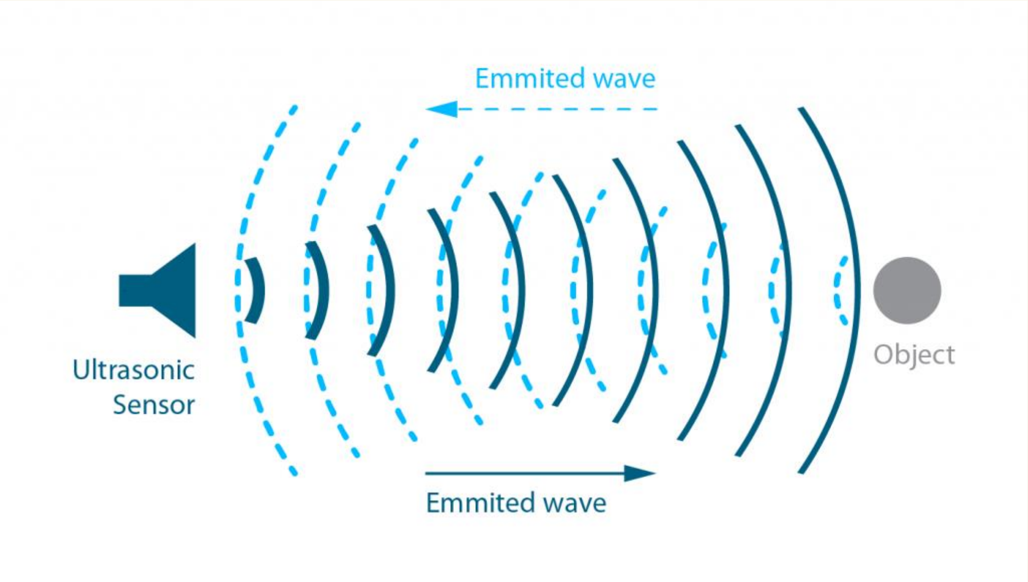
Methodology

COMPONENTS USED

- *Arduino Uno – Control unit.*
- *Sensors – Detect obstacles by measuring distance.*
- *L298 Motor Driver – Controls motor speed and direction.*
- *DC Motors – Move the car forward and backward.*



OBSTACLE DETECTION PROCESS



•*Sensors emit ultrasonic waves that reflect upon hitting an obstacle.*

•*The Arduino calculates the distance based on signal return time.*

•*If an obstacle is detected within a set range, the car stops and adjusts its path.*

CONTROL ALGORITHM

```
distance_F = Ultrasonic_read();  // Get the forward distance
if ((digitalRead(R_S) == 0) && (digitalRead(L_S) == 0)) {
  if (distance_F > Set) {    // Safe distance
    forword();               // Move forward
  } else {                   // Unsafe distance
    Check_side();            // Stop and check sides for turning
  }
}
```

•If distance > safe limit → Move forward.

•If distance < safe limit → Stop and turn.

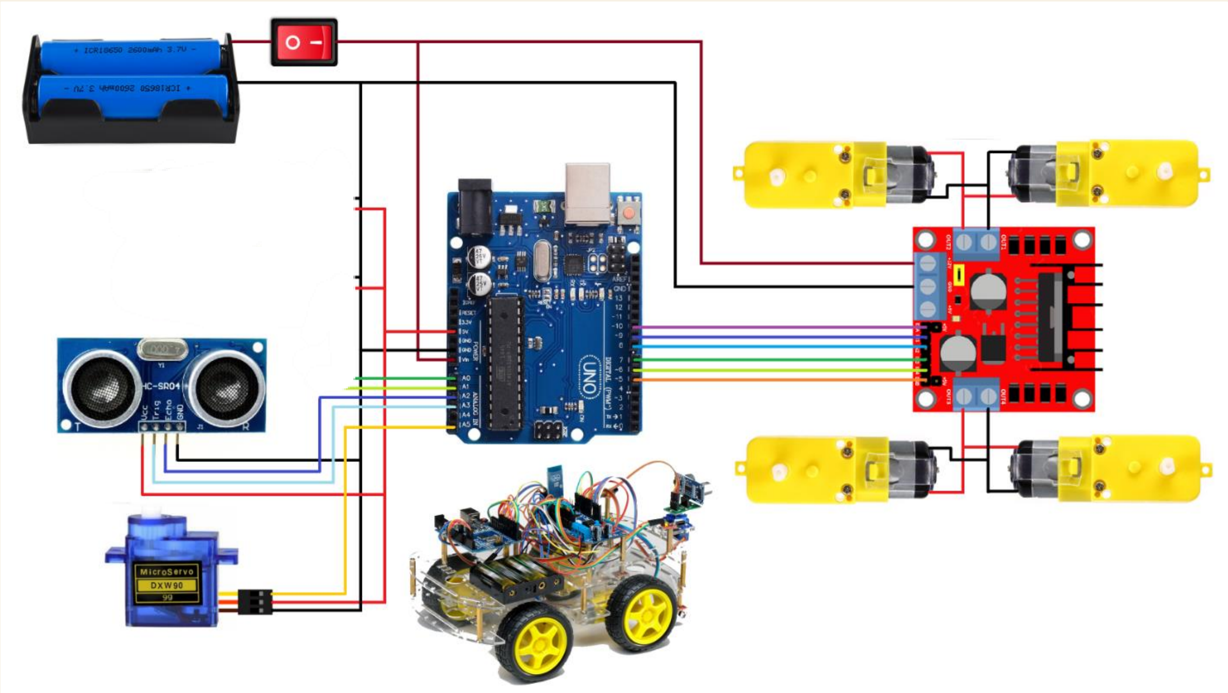
•Continuous loop for real-time detection.

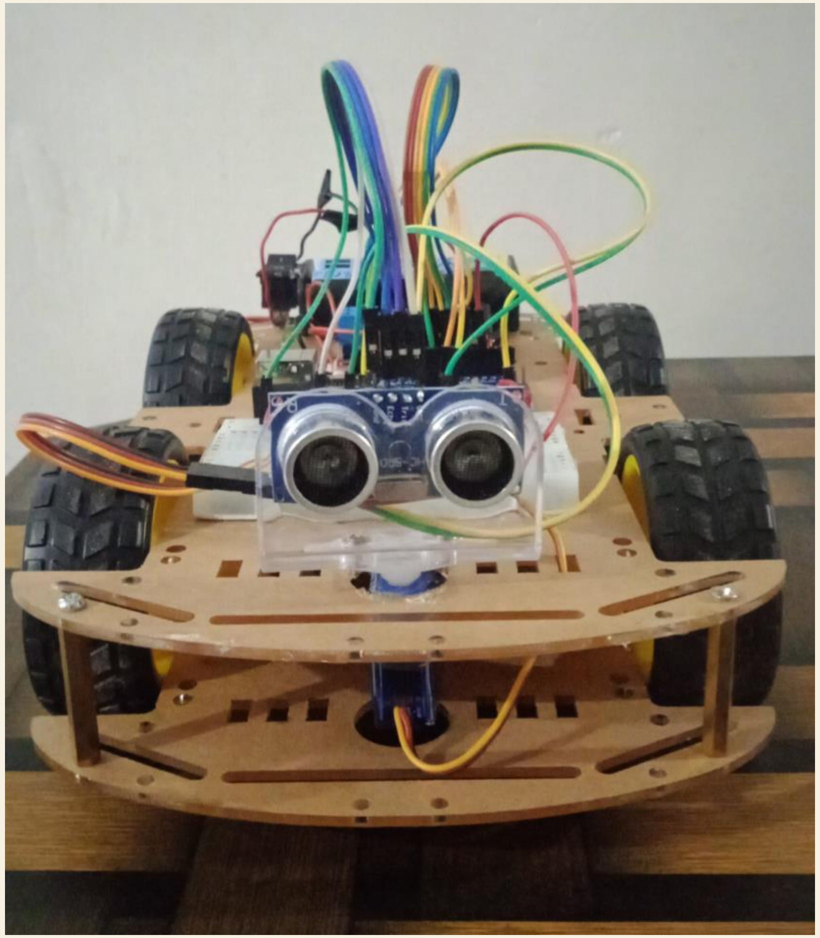
Screenshots

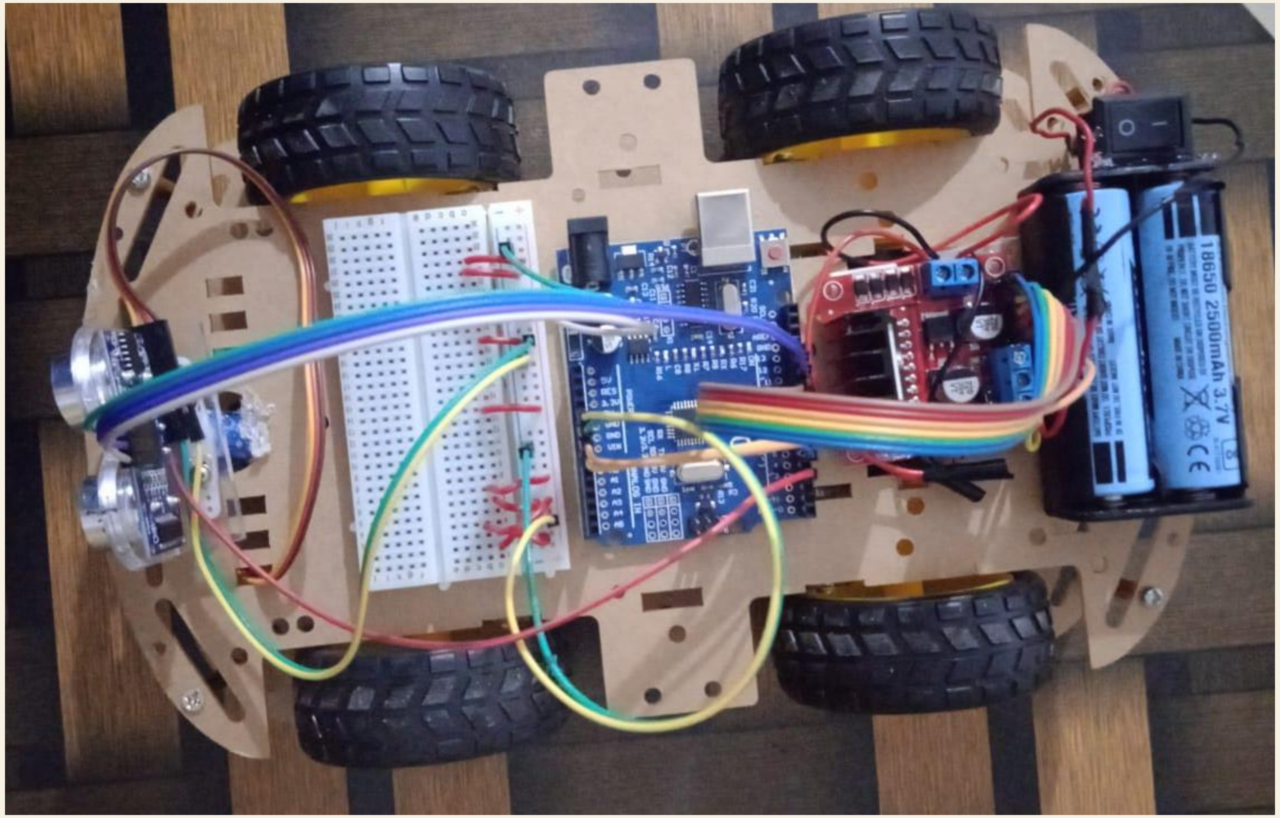
```
void forword(){  //forword
digitalWrite(in1, LOW); //Left Motor backword Pin
digitalWrite(in2, HIGH); //Left Motor forword Pin
digitalWrite(in3, HIGH); //Right Motor forword Pin
digitalWrite(in4, LOW); //Right Motor backword Pin
}

void backword(){ //backword
digitalWrite(in1, HIGH); //Left Motor backword Pin
digitalWrite(in2, LOW); //Left Motor forword Pin
digitalWrite(in3, LOW); //Right Motor forword Pin
digitalWrite(in4, HIGH); //Right Motor backword Pin
}

void turnRight(){ //turnRight
digitalWrite(in1, LOW); //Left Motor backword Pin
digitalWrite(in2, HIGH); //Left Motor forword Pin
digitalWrite(in3, LOW); //Right Motor forword Pin
digitalWrite(in4, HIGH); //Right Motor backword Pin
}
```






18650 2500mAh 3.7V

Conclusion

- The automated toy car for obstacle detection demonstrates a simplified yet effective approach to autonomous driving technology.

- By integrating ultrasonic sensors, Arduino, and motor control, the model successfully detects obstacles and adjusts its path, ensuring smooth navigation.

- This project not only highlights the potential of sensor-based systems in enhancing vehicle safety but also serves as a practical learning experience in robotics and embedded systems.

- Future improvements could include advanced path planning and object recognition to enhance the car's decision-making capabilities.

Any Questions?


Thank You!